

Sri Sivasubramaniya Nadar College of Engineering, Kalavakkam
(An Autonomous Institution, Affiliated to Anna University, Chennai)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Course Code	UCS3501	Course Name	Computer Networks						
Course Type	Theory	Course Category	Professional Core (PC)	L	T	P	TW&SL	TH	C
				45	0	0	45	90	3
Regulation	2024		Academic Year & Batch	2026 - 27 & 2024 - 28					
Degree and Branch			B.E. Computer Science & Engineering						
Semester			V	Section: A, B & C					
Department Offering the Course			Computer Science and Engineering						
Assignment – 2			Due Date:07-10-2026						

Instructions: Prepare and submit the answers in A4 sheet papers for manual submission. Scan the same and submit the scanned copy as a single pdf in the LMS.

Assignment:2 - Software-Defined Networking (SDN) (K6, CO5, CO6, 1.4.1, 6.2.1, 7.2.2, 7.1.1, 8.3.1, 9.1.1, 12.3.1, 12.4.1) (WP1, WP2, WP3) (SDG 9.4 & 12.5) (WK3, WK4, WK5, WK8, EA5)

1. Design and implement an SDN-based network architecture for a smart campus to provide centralized network management and efficient traffic forwarding. K6, CO5, CO6

Tasks:

- i. Design an SDN topology consisting of an SDN controller, OpenFlow switches, and hosts representing different campus departments using **Mininet** or an equivalent SDN emulator. **(15 Marks)**
- ii. Configure flow rules to enable communication between hosts while implementing traffic isolation between selected departments. **(15 Marks)**
- iii. Evaluate the designed network based on scalability, manageability, and communication efficiency, and document the implementation with appropriate topology diagrams and observations. **(10 Marks)**

2. Develop an SDN solution to optimize network traffic under varying communication loads. K6, CO5, CO6

Tasks:

- i. Create multiple traffic scenarios using an SDN emulator and implement suitable flow rules to optimize packet forwarding. **(15 Marks)**
- ii. Compare the network performance before and after optimization using parameters such as throughput, latency, packet loss, and flow setup time. **(15 Marks)**
- iii. Present the implementation, performance analysis, and justification for the selected optimization strategy in a comprehensive report. **(10 Marks)**

3. Design and implement a dynamic routing policy using an SDN controller for different network conditions. K6, CO5, CO6

Tasks:

- i. Develop an SDN topology with multiple communication paths between hosts and configure an SDN controller to install appropriate forwarding rules. **(15 Marks)**
- ii. Simulate link failure or congestion and modify the routing policy to automatically redirect traffic through an alternate path. **(15 Marks)**
- iii. Analyze the effectiveness of the implemented routing policy and justify its suitability based on network performance metrics. **(10 Marks)**

4. Create a secure SDN environment by implementing controller-based access control and traffic filtering mechanisms. K6, CO5, CO6

Tasks:

- i. Design an SDN network where only authorized hosts are permitted to access selected network services through controller-defined flow rules. **(15 Marks)**
- ii. Implement traffic filtering policies to block unauthorized communication and verify their effectiveness through simulation. **(15 Marks)**
- iii. Evaluate the security mechanisms implemented and recommend additional measures to improve the resilience of the SDN network. **(10 Marks)**

5. Develop an SDN-based network management solution for a real-world application and validate its performance through simulation. K6, CO5, CO6

Tasks:

- i. Select a real-world application (e.g., Smart Campus, Smart Hospital, Data Center, Smart Factory, or Enterprise Network) and design a suitable SDN architecture to meet its networking requirements. **(15 Marks)**
- ii. Implement the proposed architecture using **Mininet** (or an equivalent SDN emulator) and configure the controller to manage traffic dynamically according to the application requirements. **(15 Marks)**
- iii. Analyze the simulation results, evaluate the effectiveness of the proposed solution based on network performance and scalability, and prepare a comprehensive technical report with conclusions and future enhancements. **(10 Marks)**

Evaluation Criteria:

S.No	Question	Marks	KL	CO	PI
1.	Design a suitable SDN network for the real time application.	15	K6	CO6	1.4.1, 6.2.1, 7.2.2,
2.	Implementing the SDN architecture for the proposed scenario.	15	K4	CO6	8.3.1, 9.1.1
3.	Create a technical report based on the design and implementation of the solution.	10	K6	CO6	12.3.1, 12.4.1

Prepared By	Reviewed By	Approved By
Course Coordinator	PAC Team	HOD